# Computing Mathematics

| | Normative 4-year Degree |
|---|---|
| **Major** | Computing Mathematics<br>計算數學 |
| **Degree / Award Title** | Bachelor of Science in Computing Mathematics<br>理學士(計算數學) |
| **Offering Academic Unit** | Department of Mathematics |
| **Normal Period of Study** | 4 years |
| **Maximum Period of Study** | 8 years |
| **Credit Units Required for Graduation** | 122 credit units |

*Note: The following curriculum information is subject to periodic review and changes.*

## GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

| | | Credit Units |
|---|---|---|
| **GE University Requirements** | GE1401 University English | 3 |
| | GE2401 English for Science | 3 |
| | GE1501 Chinese Civilisation - History and Philosophy | 3 |
| | GE1601 Whole-Person Development | 1 |
| **GE Distributional Requirements** | Take at least 1 course from each of the three distributional areas:<br><br>Area 1: Arts and Humanities<br>Area 2: Study of Societies, Social and Business Organisations<br>Area 3: Science and Technology<br><br>Visit the GE website for more course information. | 12 |

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| **College-specified Courses** *(including GREAT Stream)* | MA1200 Calculus and Basic Linear Algebra I<br>*Or*<br>MA1300 Enhanced Calculus and Linear Algebra I | 3 |
| | MA1201 Calculus and Basic Linear Algebra II<br>*Or*<br>MA1301 Enhanced Calculus and Linear Algebra II | 3 |
| | CS1302 Introduction to Computer Programming | 3 |

## College / School Requirements (6-13 credit units) Catalogue Term : Semester A 2025/26

***(i) For taking the Majors without Stream (6 credit units)***

a. Choose **two** from the following four science courses subject areas:

*Biology*

| Course Code | Course Title | Credit Units |
|---|---|---|
| CHEM1200 | Discovery in Biology | 3 |

*Chemistry*

| Course Code | Course Title | Credit Units |
|---|---|---|
| CHEM1101 | Introduction to Chemistry | 3 |
| CHEM1300 | Principles of General Chemistry | 3 |

*Mathematics*

| Course Code | Course Title | Credit Units |
|---|---|---|
| MA1501 | Coordinate Geometry | 3 |
| MA1502 | Algebra | 3 |

*Physics*

| Course Code | Course Title | Credit Units |
|---|---|---|
| PHY1101 | Introductory Classical Mechanics | 3 |
| PHY1201 | General Physics I | 3 |

b. **Compulsory attendance** for the following two soft skills courses:

| Course Code | Course Title | Credit Units |
|---|---|---|
| CSCI1001 | Employability for Scientists | 0 |
| CSCI1002 | Career Lab for Scientists | 0 |

***(ii) For taking the Stream of "Global Research Enrichment and Technopreneurship Programme (GREAT)" (13 credit units)***

a. Research Methodology

| Course Code | Course Title | Credit Units |
|---|---|---|
| CSCI2002 | Workshop on Research Methodology | 1 |

b. College Requirements I (6 credit units). Choose **two** from the following six courses:

*Chemistry*

| Course Code | Course Title | Credit Units |
|---|---|---|
| CHEM1300 | Principles of General Chemistry | 3 |
| CHEM2006A | Principles of Inorganic Chemistry | 3 |
| CHEM2007A | Principles of Organic Chemistry | 3 |

*Mathematics*

| Course Code | Course Title | Credit Units |
|---|---|---|
| MA2509 | Discrete Mathematics | 3 |

*Physics*

| Course Code | Course Title | Credit Units |
|---|---|---|
| PHY1101 | Introductory Classical Mechanics | 3 |
| PHY1202 | General Physics II | 3 |

c. College Requirements II (6 credit units). Choose **two** from the following four courses:

*Chemistry*

| Course Code | Course Title | Credit Units |
|---|---|---|
| CHEM2004A | Principles of Analytical Chemistry | 3 |
| CHEM2008A | Principles of Physical Chemistry | 3 |

*Mathematics*

| Course Code | Course Title | Credit Units |
|---|---|---|
| MA1501 | Coordinate Geometry | 3 |

*Physics*

| Course Code | Course Title | Credit Units |
|---|---|---|
| PHY1203 | General Physics III | 3 |

## Major Requirements (63-65 credit units) Catalogue Term : Semester A 2025/26

**1. Core Courses (45 credit units)**

| Course Code | Course Title | Credit Units | Remarks |
|---|---|---|---|
| MA2503 | Linear Algebra | 4 | * |
| MA2507 | Computing Mathematics Laboratory | 1 | Waived for students admitted into ASII |
| MA2508 | Multi-variable Calculus | 4 | |
| MA2509 | Discrete Mathematics | 3 | * |
| MA2510 | Probability and Statistics | 3 | * |
| MA3511 | Ordinary Differential Equations | 3 | |

| | | | |
|---|---|---|---|
| MA3512 | Partial Differential Equations | 3 | |
| MA3514 | Numerical Methods for Differential Equations | 3 | |
| MA3515 | Introduction to Optimization | 3 | |
| MA3517 | Complex Analysis | 3 | |
| MA3518 | Applied Statistics | 3 | |
| MA3525 | Elementary Numerical Methods | 3 | |
| MA3526 | Analysis | 3 | |
| CS2360 | Java Programming | 3 | |
| CS2468 | Data Structures and Data Management | 3 | |

*Note: * Based on the academic background, two out of the three core courses MA2503, MA2509 and MA2510 are to be waived for students admitted with Advanced Standing II.*

**2. Electives (18 - 20 credit units)**

***For Normative 4-year, please choose one of the following courses (3 - 8 credit units):***

| Course Code | Course Title | Credit Units | Remarks |
|---|---|---|---|
| MA4530 | Project | 6 | |
| MA4555 | Supervised Individual Research | 3 | |
| CSCI4001 | Co-operative Education Scheme for Science Students | 6 | |
| CSCI4002 | Industrial Attachment Scheme for Science Students | 3 | |
| CSCI4005 | Overseas Internship Scheme for Science Student | 3 | |

***Fulfill the Major Electives requirement from the following list***
***Normative 4-year: remaining 10 - 17 credit units***

| Course Code | Course Title | Credit Units | Remarks |
|---|---|---|---|
| MA1501 | Coordinate Geometry | 3 | Only for first year students of Normative 4-year Degree (BSCSIU4/CM) except with special approval. |
| MA3521 | Introductory Mathematical Finance | 3 | |
| MA3523 | Introduction to Abstract Algebra | 3 | |
| MA3530 | Selected Topics on Mathematics | 1 | |
| MA4523 | Introduction to Finite Element Method | 3 | |
| MA4524 | Elementary Number Theory and Applications | 3 | |

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|---|---|---|---|
| MA4525 | Combinatorial and Network Optimization | 3 | |
| MA4527 | Computational Geometry | 3 | |
| MA4528 | Introduction to Dynamical Systems and Chaos | 3 | |
| MA4529 | Mathematical Finance | 3 | |
| MA4530 | Project | 6 | |
| MA4531 | Partial Differential Equations II | 3 | |
| MA4532 | Applied Functional Analysis | 3 | |
| MA4533 | Applied Mathematics Laboratory | 1 | |
| MA4534 | Computer Graphics and Geometry | 3 | |
| MA4535 | Applied Probability | 3 | |
| MA4537 | Introduction to Actuarial Science | 3 | |
| MA4538 | Numerical Partial Differential Equations | 3 | |
| MA4540 | Modelling and Case Studies | 3 | |
| MA4542 | Real Analysis | 3 | |
| MA4543 | Introduction to Time Series and Forecasting | 3 | |
| MA4545 | Applied Differential Geometry | 3 | |
| MA4546 | Introduction to Stochastic Processes | 3 | |
| MA4547 | Asymptotic Analysis | 3 | |
| MA4548 | Abstract Algebra II | 3 | |
| MA4549 | Sampling Survey Methods for Social and Market Research | 3 | |
| MA4550 | A Mathematical Introduction to Machine Learning for Data Sciences | 3 | |
| MA4551 | Introduction to Functional Analysis | 3 | |
| MA4552 | Introduction to Differential Manifolds | 3 | |
| MA4553 | A Mathematical Introduction to Image Processing and Analysis, with Some Surprising Applications | 3 | |
| MA4554 | Statistical Foundations of Data Science | 3 | |
| CS4486 | Artificial Intelligence | 3 | |
| CS4487 | Machine Learning | 3 | |
| CSCI3001 | Grand Challenges in the World | 3 | |

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|---|---|---|---|
| SDSC3001 | Big Data: The Arts and Science of Scaling | 3 | |
| SDSC3003 | Blockchain | 3 | |
| SDSC3022 | Financial Data Analytics for Investments | 3 | |
| SDSC4107 | Financial Engineering and Analytics | 3 | |

**Within the abovementioned credit units for the BSCM Major Electives requirement, students may declare a maximum of one of the following streams by fulfilling the designated stream requirement. Declaration of stream is optional and there is no GPA requirement for declaring a stream.**

**Enriched Mathematics Stream (12 credit units)**

| Course Code | Course Title | Credit Units | Remarks |
|---|---|---|---|
| MA3523 | Introduction to Abstract Algebra | 3 | |
| MA4542 | Real Analysis | 3 | |
| MA4545 | Applied Differential Geometry | 3 | |

***Choose at least one of the following courses:***

| Course Code | Course Title | Credit Units | Remarks |
|---|---|---|---|
| MA4528 | Introduction to Dynamical Systems and Chaos | 3 | |
| MA4546 | Introduction to Stochastic Processes | 3 | |
| MA4551 | Introduction to Functional Analysis | 3 | |

**Financial Mathematics Stream (12 credit units)**

| Course Code | Course Title | Credit Units | Remarks |
|---|---|---|---|
| MA3521 | Introductory Mathematical Finance | 3 | |
| MA4529 | Mathematical Finance | 3 | |

***Choose at least two of the following courses:***

| Course Code | Course Title | Credit Units | Remarks |
|---|---|---|---|
| MA4537 | Introduction to Actuarial Science | 3 | |
| EF3320 | Security Analysis and Portfolio Management | 3 | |
| EF4821 | Derivatives Pricing II: Interest Rate and Credit Risk | 3 | |

**3. GREAT Stream Core Courses (56 credit units)**

| Course Code | Course Title | Credit Units | Remarks |
|---|---|---|---|
| MA3510 | Independent Research I | 6 | |
| MA4510 | Independent Research II | 6 | |
| MA2503 | Linear Algebra | 4 | |
| MA2508 | Multi-variable Calculus | 4 | |
| MA2510 | Probability and Statistics | 3 | |
| MA3511 | Ordinary Differential Equations | 3 | |
| MA3512 | Partial Differential Equations | 3 | |
| MA3514 | Numerical Methods for Differential Equations | 3 | |
| MA3515 | Introduction to Optimization | 3 | |
| MA3517 | Complex Analysis | 3 | |
| MA3518 | Applied Statistics | 3 | |
| MA3525 | Elementary Numerical Methods | 3 | |
| MA3526 | Analysis | 3 | |
| CS2360 | Java Programming | 3 | |
| CS2468 | Data Structures and Data Management | 3 | |
| MGT2324 | Introduction to Entrepreneurship | 3 | |

## Suggested Study Plan

Normative 4-year Degree (BSCSIU4/CM/2025)